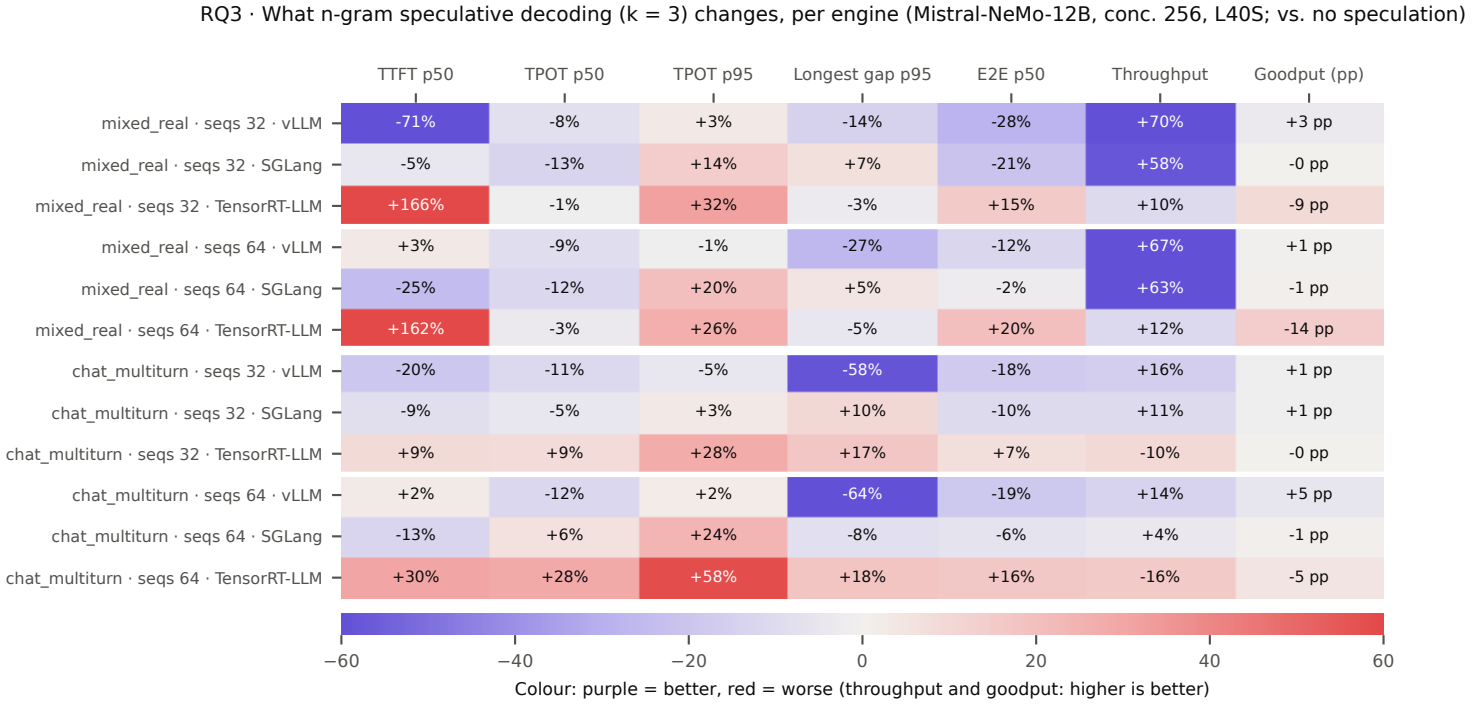
RQ3 · What n-gram speculative decoding (k = 3) changes, per engine (Mistral-NeMo-12B, conc. 256, L40S; vs. no speculation)
TTFT p50
TPOT p50
TPOT p95
Longest gap p95
E2E p50
Throughput
Goodput (pp)
mixed_real · seqs 32 · vLLM
-71%
-8%
+3%
-14%
-28%
+70%
+3 pp
mixed_real · seqs 32 · SGLang
-5%
-13%
+14%
+7%
-21%
+58%
-0 pp
mixed_real · seqs 32 · TensorRT-LLM
+166%
-1%
+32%
-3%
+15%
+10%
-9 pp
mixed_real · seqs 64 · vLLM
+3%
-9%
-1%
-27%
-12%
+67%
+1 pp
mixed_real · seqs 64 · SGLang
-25%
-12%
+20%
+5%
-2%
+63%
-1 pp
mixed_real · seqs 64 · TensorRT-LLM
+162%
-3%
+26%
-5%
+20%
+12%
-14 pp
chat_multiturn · seqs 32 · vLLM
-20%
-11%
-5%
-58%
-18%
+16%
+1 pp
chat_multiturn · seqs 32 · SGLang
-9%
-5%
+3%
+10%
-10%
+11%
+1 pp
chat_multiturn · seqs 32 · TensorRT-LLM
+9%
+9%
+28%
+17%
+7%
-10%
-0 pp
chat_multiturn · seqs 64 · vLLM
+2%
-12%
+2%
-64%
-19%
+14%
+5 pp
chat_multiturn · seqs 64 · SGLang
-13%
+6%
+24%
-8%
-6%
+4%
-1 pp
chat_multiturn · seqs 64 · TensorRT-LLM
+30%
+28%
+58%
+18%
+16%
-16%
-5 pp
−60
−40
−20
0
20
40
60
Colour: purple = better, red = worse (throughput and goodput: higher is better)